

PIPELINE STAGE SPECIFICATIONS

STAGE	MODULE / ENGINE	INPUT	OUTPUT	KEY PARAMETERS	
1. Input Reception	Web / Mobile API Gateway	HTTP req · voice blob	Normalised query string	lang_hint · session_id	
2. Language ID	Statistical LID Classifier	Raw query string	lang_code ∈ {en, as}	conf_threshold = 0.85	
3. Query Embedding	BGE-M3 (1024-dim)	Query string	float[1024] vector	model=bge-m3; pool=cls	
4. Topic Matching	Cosine Similarity Engine	Query vector + topic DB	topic_id, score	n=197; threshold=0.72	
5. Scope Resolution	Curriculum Resolver Svc	topic_id	chapter-subject-board	hierarchy_depth = 6	
6. RAG Retrieval	HNSW Vector Index	Query vec + filters	Top-K text chunks	k=5; ef_search=128	
7. Context Assembly	Prompt Builder	chunks + memory + scope	Structured prompt string	max_ctx = 12 000 tokens	
8. LLM Generation	Gemini Flash / Sarvam 30B	Prompt string	Answer text stream	max_new=800; temp=0.3	
9. Response Delivery	SSE Stream + TTS Engine	Answer text	JSON stream + audio	format=SSE; voice=indic	
10. Memory Write	Student Profile Store	Session summary	Updated memory doc	TTL=∞; upsert=true	

TECHNICAL DIFFERENTIATORS

01 · Dual-Language RAG Pipeline

Single retrieval pipeline serves both English and Assamese. LLM handles generation language without separate indices, saving space for both languages. Metadata filter applied to both.

02 · Curriculum-Scoped Retrieval

Metadata filters applied at vector-search time – chunks are syllabus-relevant, reducing hallucination. Supports groups × N subjects.

03 · Persistent Student Memory

MongoDB document per student accumulates context across sessions. Prior-context block (max 800 tokens). Enables resume. Writes after each session using upsert.

04 · Human-in-the-Loop Content Quality Control

Staff publish pipeline: raw_content → chunker → LLM → answer. Version-controlled with optimistic locking (version increments on update). Human review supported at any stage.

05 · Edge-First Architecture

Cloudflare Workers handle routing, JWT auth, and cache invalidation. Edge caches for cached responses. LLM origin (Cloud Run) distributed KV. No cold-start latency on auth or routing.

SYSTEM CONSTRAINTS & PERFORMANCE TARGETS

Latency

TTFB <400ms (edge)
LLM stream start <2s

Throughput

1,000 concurrent users
Stateless horizontal scale

RAG Accuracy

In-scope hallucination rate <2%

Language Support

English (en) + Assamese (as)
Extensible to 22 IN langs

Content Coverage

SEBA Cl.9-10 · AHSEC Cl.11-12
Core subjects + PYQ archive

Auth & Security

JWT RS256 · rate limiting
HTTPS-only · CORS enforced